

Linked lists

C programming · Structures and dynamic memory

ISTIC · University of Rennes 2026-2027



Kahoot corrections

Dynamic memory allocation · q1-q10

Kahoot (alloc) q1: pointer errors

```
/* X */ m = malloc(5); m = NULL;  
/* Y */ free(n); n->value = 5;  
/* Z */ char *p; *p = 'a';
```

1. Using dangling pointers
2. Using uninitialised pointers
3. Lost memory

Match X, Y and Z to the problems.

- A. X-1, Y-3, Z-2
- B. X-3, Y-2, Z-1
- C. X-2, Y-1, Z-3
- D. **X-3, Y-1, Z-2**

- X leaks if allocation succeeds; its address is lost.
- Y uses a freed object; Z has no initialised address.

Kahoot (alloc) q2: *free(NULL)*

```
#include <stdio.h>
int main() {
    int *p = (int *)malloc(sizeof(int));
    p = NULL;
    free(p);
}
```

- *free(NULL)* does nothing (CM, slide 291).
- D assumes allocation succeeds. No alias is saved here.
- Missing *<stdlib.h>* is a separate C17 error.

What is the problem with this code?

- A. Compiler error: *free()* can't be applied on NULL pointer
- B. Dangling pointer
- C. The program may crash as *free()* is called for NULL pointer.
- D. **Memory leak**

Kahoot (alloc) q2: another reference

```
int *p = malloc(sizeof(int));  
if (p == NULL) return EXIT_FAILURE;  
int *q = p;  
p = NULL;  
free(p);  
free(q);
```

- *q* kept the allocated address before *p* changed.
- *free(q)* releases the block. **No leak here.**

Kahoot (alloc) q3: return value of *free()*

```
#include <stdio.h>
#include <stdlib.h>
int main() {
    int *ptr1, *ptr2;
    ptr1 = malloc(4);
    *ptr1 = 10;
    *ptr2 = free(ptr1);
    printf("%d\n", *ptr2);
    return 0;
}
```

What is the output?

- A. 10
 - B. The address stored in ptr1
 - C. The address stored in ptr2
 - D. **Error**
- *free()* returns *void*: no value to assign.
 - *ptr2* is also uninitialised.

Kahoot (alloc) q4: a local address

```
#include <stdio.h>
int *fun() {
    int x = 5;
    return &x;
}
int main() {
    int *p = fun();
    printf("%d", *p);
    return 0;
}
```

What is the problem with this code?

- A. Memory leak
- B. **Dangling pointer**
- C. There is no problem
- D. Compilation error

- x stops existing when $fun()$ returns.
- Dereferencing p has **undefined behaviour**.

Kahoot (alloc) q5: one block

```
#include <stdio.h>
#include <stdlib.h>
int main() {
    int r = 3, c = 4;
    int *arr = (int *)malloc(
        r * c * sizeof(int));
    int i, j, count = 0;
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            ??? = ++count;
    /* ... */
    return 0;
}
```

Which expression replaces ????

A. **(arr + i*c + j)*

B. *arr[i][j]*

C. *arr + i*c + j*

D. **(&arr[i]+&arr[j])*

- Skip *i* rows of *c* integers, then *j* integers.
- Also written *arr[i*c + j]*.

Kahoot (alloc) q6: separate rows

```
#include <stdio.h>
#include <stdlib.h>
int main() {
    int r = 3, c = 4, i, j, count;
    int *arr[r];
    for (i = 0; i < r; i++)
        arr[i] = (int *)malloc(
            c * sizeof(int));
    count = 0;
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            ??? = ++count;
    /* ... */
    return 0;
}
```

Which expression replaces ?????

- A. $*(arr+i*c+j)$
- B. $((*(arr + i*c) + j)$
- C. $((*(arr + i*c + j))$
- D. $((***(arr+i)**+j)$

- $*(arr+i)$ selects row i ; $+j$ selects column j .
- Also written $arr[i][j]$ or $((*(arr+i))[j]$.

Kahoot (alloc) q7: freeing the grid

```
#include <stdio.h>
#include <stdlib.h>
int main() {
    int r = 3, c = 4, i, j, count;
    int **arr = (int **)malloc(
        r * sizeof(int *));
    for (i = 0; i < r; i++)
        arr[i] = (int *)malloc(
            c * sizeof(int));
    count = 0;
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            arr[i][j] = ++count;
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            printf("%d ", arr[i][j]);
    free(arr);
    return 0;
}
```

Is this code correct?

- A. Yes
- B. **No**
- C. I do not know

- `free(arr)` releases only the pointer table.
- On success, **the three rows leak**. Free them first:

```
for (i = 0; i < r; i++)
    free(arr[i]);
free(arr);
```

Kahoot (alloc) q8: initialisation

```
struct Point {  
    int x = 0;  
    int y = 0;  
};
```

What is the output? (C17)

- A. x=0, y=0
 - B. **Compilation error**
 - C. Nothing of the mentioned
-
- A C structure definition cannot initialise its members.
 - Initialise an **object** after defining its type.

Kahoot (alloc) q9: structure copy

```
#include <stdio.h>
struct student { char *name; };
struct student s;
struct student fun(void) {
    s.name = "newton";
    printf("%s\n", s.name);
    s.name = "alan";
    return s;
}
void main() {
    struct student m = fun();
    printf("%s\n", m.name);
    m.name = "turing";
    printf("%s\n", s.name);
}
```

What is the output?

- A. **newton · alan · alan**
 - B. alan · newton · alan
 - C. newton · alan · turing
 - D. Compilation error
- Assigning *m.name* does not change *s.name*.
 - A assumes *void main()* is accepted; strict compilation can reject it. Use *int main(void)*.

Kahoot (alloc) q9: the caller

```
struct student m = fun();  
printf("%s\n", m.name);  
m.name = "turing";  
printf("%s\n", s.name);
```

Pointer value	After <i>fun()</i>	After assignment
<i>m.name</i> points to	"alan"	"turing"
<i>s.name</i> points to	"alan"	"alan"

- Returning *s* copies its pointer member into *m*.
- The last *printf()* reads *s.name*, still **alan**.

Kahoot (alloc) q10: member access

```
#include <stdio.h>
struct point { int x; int y; };
void foo(struct point *);
int main() {
    struct point p1 = {1, 2};
    foo(&p1);
}
void foo(struct point *p) {
    printf("%d\n", *p.x++);
}
```

What is the output?

A. **Compile time error**

B. Segmentation fault

C. 2

D. 1

- `.` binds before `*`, but `p` is a pointer.
- `p->x++` or `(*p).x++` would print **1**, then set `x` to **2**.

Struct reminder

Each list node will hold a song ID and a pointer.

A structure groups them together.

Struct: type and object

```
struct Point {  
    int x;  
    int y;  
};  
struct Point origin = {0, 0};  
origin.x = 3;
```

- ***struct Point* describes the type; *origin* is an object.**
- *origin.x* accesses that object's x member.

Typedef: a type alias

```
typedef struct Point point_t;
```

```
struct Point a = {1, 2};
```

```
point_t b = {1, 2};
```

- *struct Point* and *point_t* name the **same type**.
- *typedef* creates no object and allocates no heap memory.

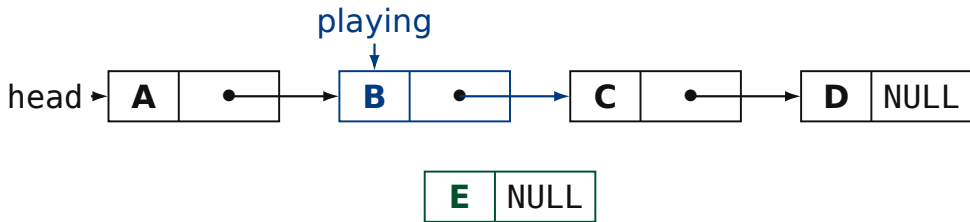
1. A playlist that changes

A listener requests a song next.

Can we insert it without shifting the rest of the playlist?

Play E immediately after B

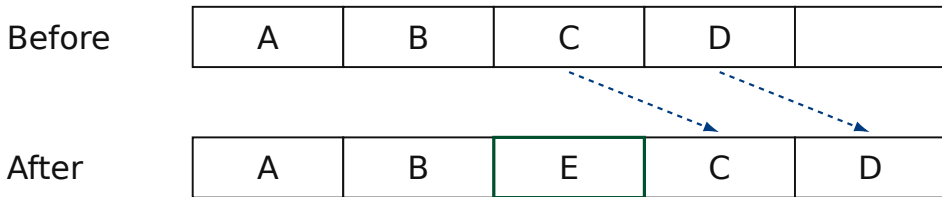
- **B is playing.** A listener requests **E next.**



- We want **A → B → E → C → D**. Keep C and D in the playlist.

With an array, make room for E

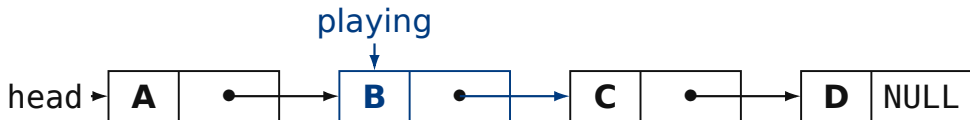
- The array has a spare slot. Where do C and D go?



- C and D move one slot.** A longer suffix means more moves.

A node stores a song and a link

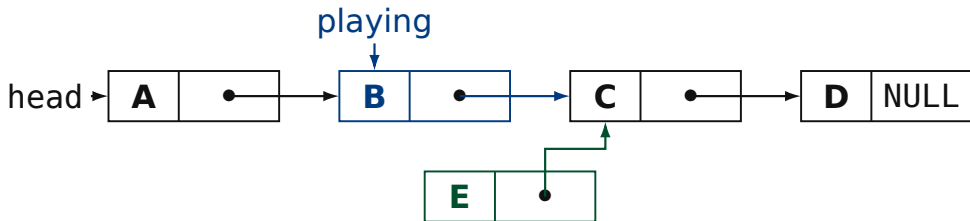
- A node stores a song ID and the address of the next node.



- The dot is a pointer value; the arrow leads to its node. **NULL ends the list.**

First, let E remember C

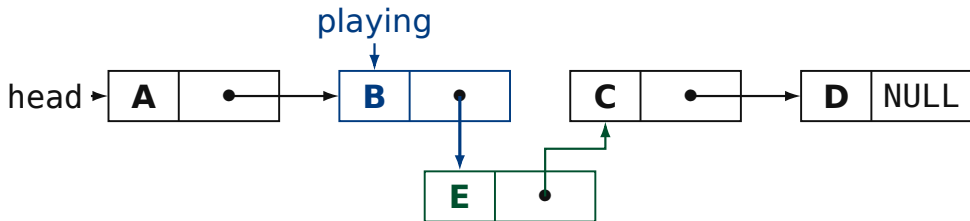
- Before changing B, keep its current successor.



- E → C**. Following the playlist from *head* still gives A B C D.

Then, make B lead to E

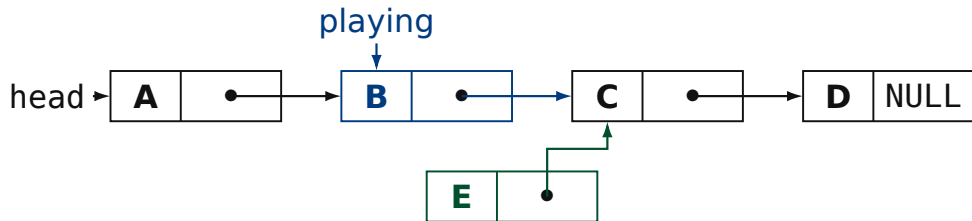
- Change the next-song link stored in B.



- Follow the arrows: **A → B → E → C → D**. **Two links changed.**

Cancel the request

- E has not played yet. Make B lead to C again.



- E is **unlinked**. Its storage still needs to be released.

When links help

- We already know the current song: **B**.
- Adding or removing its successor changes a few links.
- **The other nodes stay in place.**
- Finding song **number 500** requires **following the chain**.
- Arrays give direct indexing and store elements together.

The same edit with numbers

Insert **101** into $[100, 103, 105, 200]$. What must change?

- **Array with capacity:** move 103, 105 and 200 one slot.
- **List:** find 100, then link **100** $\rightarrow$ **101** $\rightarrow$ **103**.
- The list needs an extra pointer per node and sequential search.

2. Linked-list basics

Turn the drawn playlist into a working C program: **A → B → NULL.**

- Allocate two nodes.
- Link them using *next*.
- Walk the list without changing *head*.

Can we use the alias here?

```
typedef struct list_elem {  
    int value;  
    list_elem_t *next;  
} list_elem_t;
```

- At *next*, does the compiler already know *list_elem_t*?
- GCC: **unknown type name** '*list_elem_t*'.
- This typedef declares the alias **after the closing brace**.

A tag inside, an alias afterwards

```
typedef struct list_elem {  
    int value;  
    struct list_elem *next;  
} list_elem_t;
```

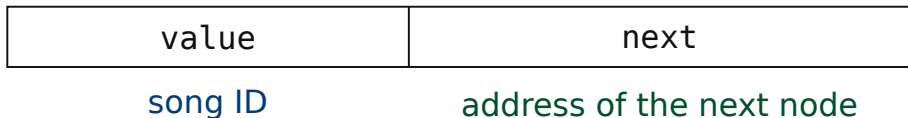
- *struct list_elem* is **already known inside the braces**.
- *next* stores a pointer. The node's full size is not needed yet.
- After this declaration, *list_elem_t* names **the same type**.

Now we can use the alias:

```
list_elem_t *head = NULL;
```

What is in one node?

- A song ID and the address of the next node.



- The next node is a **separate object**, possibly far away in memory.
- *next* stores its address; the nodes need not be adjacent.

Start with an empty playlist

```
list_elem_t *head = NULL;
```

- *head* is a **local pointer** in *main()*.
- Its value is *NULL*: there is no first node yet.
- No *malloc()* call, so **no heap node**.

0xffff **Stack (main)**

0xffff8	head	NULL
		⋮
Heap		
		⋮

0x0000

Allocate the first song

```
list_elem_t *node =  
    malloc(sizeof(list_elem_t));  
if (node == NULL)  
    return EXIT_FAILURE;  
node->value = 'A';  
node->next = NULL;  
head = node;
```

- *malloc()* creates the heap object.
- *head = node* copies its address.

0xffff **Stack (main)**

0xffff8	head	0x1000
0xffff0	node	0x1000

Heap

0x1008	next	NULL
0x1000	value	'A'

0x0000

Add a second song

```
node =  
    malloc(sizeof(list_elem_t));  
if (node == NULL) {  
    free(head);  
    return EXIT_FAILURE;  
}  
node->value = 'B';  
node->next = NULL;  
head->next = node;
```

- A and B are separate allocations.
- A's *next* stores B's address.

0xffff **Stack (main)**

0xffff8	head	0x1000
0xffff0	node	0x3000
	:	

Heap

0x3008	next	NULL
0x3000	value	'B'
	:	

0x1008	next	0x3000
0x1000	value	'A'

0x0000

Read the playlist

```
list_elem_t *p = head;
while (p != NULL) {
    printf("%c -> ", p->value);
    p = p->next;
}
puts("NULL");
```

- Only *p* advances.
- *head* still reaches the whole list.

0xffff Stack (main)

0xffff8	head	0x1000
0xffff0	node	0x3000
0xffe8	p	NULL

Heap

0x3008	next	NULL
0x3000	value	'B'

	:	
--	---	--

0x1008	next	0x3000
0x1000	value	'A'

0x0000

3. Create nodes in a function

Creating each song repeats the same allocation code.
Let *create_element()* do that work for us.

Create a node in a function

What survives when this function returns?

```
list_elem_t *create_element(int value) {  
    list_elem_t *node = malloc(sizeof(list_elem_t));  
    if (node == NULL) return NULL;  
    node->value = value;  
    node->next = NULL;  
    return node;  
}
```

In *main()*: `list_elem_t *node = create_element('A');`

Call *create_element('A')*

```
/* main pauses here */  
list_elem_t *node = create_element('A');
```

Stack: local variables

create_element() (callee)

value

'A'

main() (caller)

head

NULL

node

waiting

Heap

No node
allocated yet

- main()* waits. The called function gets **its own stack frame**.

malloc() reserves the node

```
list_elem_t *node = malloc(sizeof(list_elem_t));  
if (node == NULL) return NULL;
```

Stack: local variables

create_element() (callee)

value
node

'A'
0x1000

main() (caller)

head
node

NULL
waiting

Heap

0x1000

value	?
next	?

- Here *malloc()* returns **0x1000**. ? means uninitialised.

Initialise *value* and *next*

```
node->value = value;  
node->next = NULL;
```

Stack: local variables

`create_element()` (callee)

value
node

'A'
0x1000

`main()` (caller)

head
node

NULL
waiting

Heap

0x1000

value	A
next	NULL

- > follows *node* to write **the heap structure**.

Return the address to *main()*

```
/* inside create_element */
return node;
```

Stack: local variables

`create_element()` (callee)

value
node

'A'
0x1000

`main()` (caller)

head
node

NULL
0x1000

Heap

0x1000

value	A
next	NULL

- The return value copies **0x1000** into *main()*'s *node*.

The call ends; the node remains

```
/* back in main: the call has completed */  
list_elem_t *node = create_element('A');
```

Stack: local variables

`create_element()`: call finished

`main()` (caller)

head

NULL

node

0x1000

Heap

0x1000

value	A
next	NULL

- The local pointer is gone; **the heap node stays allocated.**

Make the node the first one

```
/* in main */  
head = node;
```

Stack: local variables

create_element(): call finished

main() (caller)

head
node

0x1000
0x1000

Heap

0x1000

value	A
next	NULL

- Two pointers hold **the same address**. There is still one node.

4. Change the caller's head

We want *insert_head(...)* to put a new song first.
How can it update *head* in *main()*?

Can the function change *head*?

Will the caller's *head* change?

```
int insert_head(list_elem_t *l, int value) {  
    list_elem_t *node = create_element(value);  
    if (node == NULL) return -1;  
    node->next = l;  
    l = node;  
    return 0;  
}
```

Call: *insert_head(head, 'A')*.

The argument is copied into *l*

```
insert_head(head, 'A');  
/* parameter: list_elem_t *l */
```

Stack: local variables

`insert_head()` (callee)

<code>l</code>	NULL
<code>value</code>	'A'

`main()` (caller)

<code>head</code>	NULL
-------------------	------

Heap

No node
allocated yet

- `l` gets **NULL**, the value of `head`. The variables are separate.

The function creates a node

```
list_elem_t *node = create_element(value);
node->next = l;
```

Stack: local variables

insert_head() (callee)

l	NULL
value	'A'
node	0x1000

main() (caller)

head	NULL
------	------

Heap

0x1000

value	A
next	NULL

- node->next* gets NULL. **head is still NULL.**

l = node changes only l

l = node;

Stack: local variables

insert_head() (callee)

l	0x1000
value	'A'
node	0x1000

main() (caller)

head	NULL
------	------

Heap

0x1000

value	A
next	NULL

- l changes.** *head* is a separate variable and stays NULL.

Return loses the node's address

```
/* inside insert_head */  
return 0;
```

Stack: local variables

insert_head(): call finished

main() (caller)

head

NULL

Heap

0x1000

value	A
next	NULL

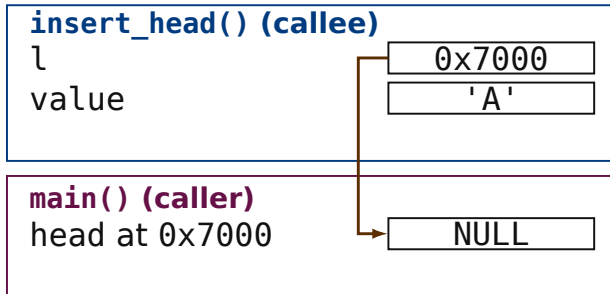
Unreachable

- The local pointers are gone; **the node is unreachable.**

Pass the address of *head*

```
insert_head(&head, 'A');  
/* parameter: list_elem_t **l */
```

Stack: local variables



Heap

No node
allocated yet

- Here *&head* is 0x7000. ****l* refers to head itself.**

Read the old *head* through **l*

```
list_elem_t *node = create_element(value);  
node->next = *l;
```

Stack: local variables

`insert_head()` (callee)

`l`

`value`

`node`

0x7000

'A'

0x1000

`main()` (caller)

`head` at 0x7000

NULL

Heap

0x1000

value	A
next	NULL

- Read `*l`: NULL, the current value stored in *head*.

Write *head* through *l

```
*l = node;
```

Stack: local variables

insert_head() (callee)

l

value

node

0x7000

'A'

0x1000

main() (caller)

head at 0x7000

0x1000

Heap

0x1000

value	A
next	NULL

- head becomes 0x1000.** l still contains &head.

Back in *main()*: *head* is updated

```
/* insert_head has returned */
printf("head -> %c -> NULL\n", head->value);
```

Stack: local variables

insert_head(): call finished

main() (caller)

head at 0x7000

0x1000

Heap

0x1000

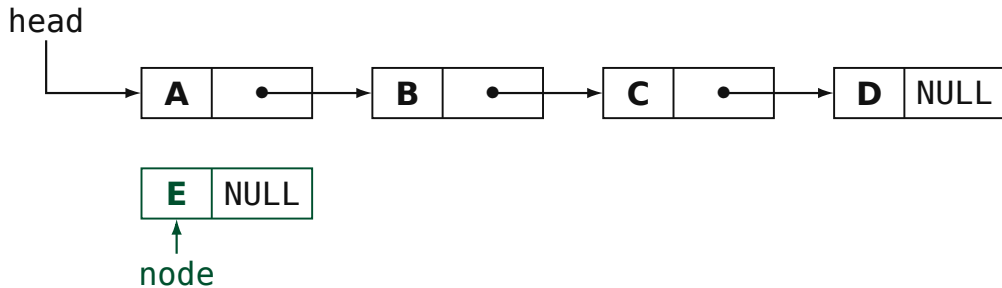
value	A
next	NULL

- C still passes by value.** The copied value was *&head*.

5. Insert without losing links

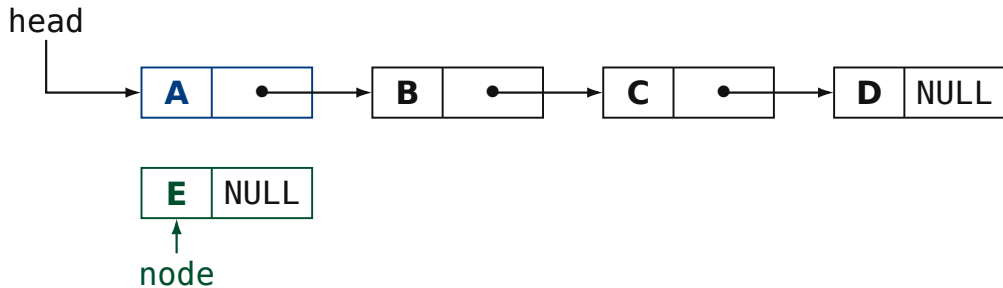
A new song must fit before, between, or after the existing songs.
Which links need to change?

Insert at the head · starting state



- $A \rightarrow B \rightarrow C \rightarrow D$. A newly allocated node **E** is ready to link.

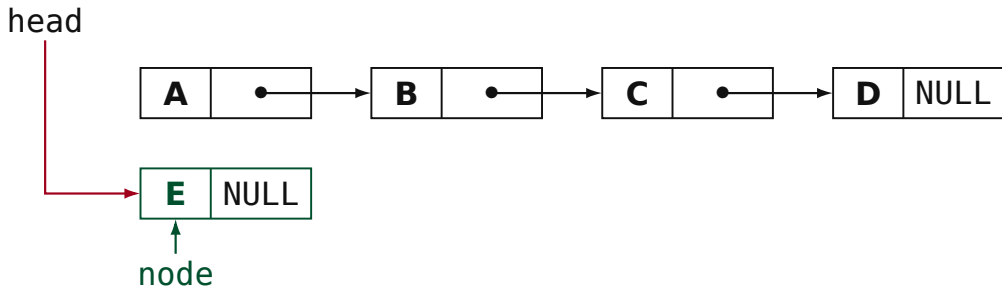
Insert at the head · which link matters?



- *head* is our entry point to the existing chain.

If we change *head* first

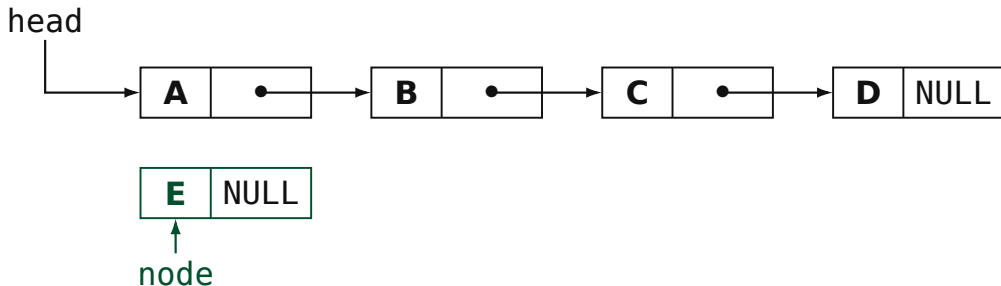
What remains reachable after $head = node$?



`head = node;`

- Only E remains reachable; **A-D leak**.

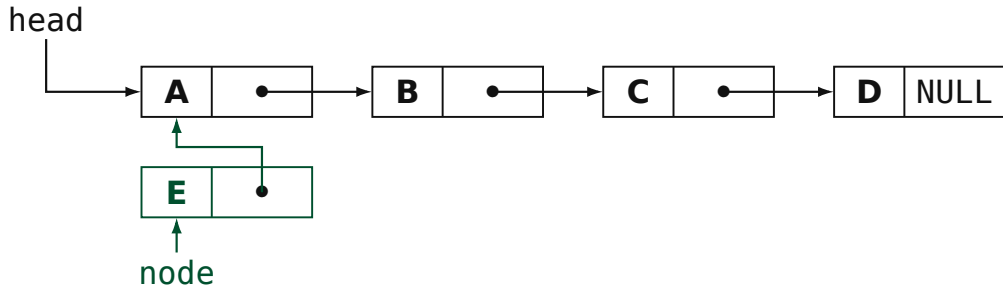
Back to the initial list



- Reset to the initial state. Preserve the old address before changing *head*.

Link E to A

Where must E point before we change *head*?

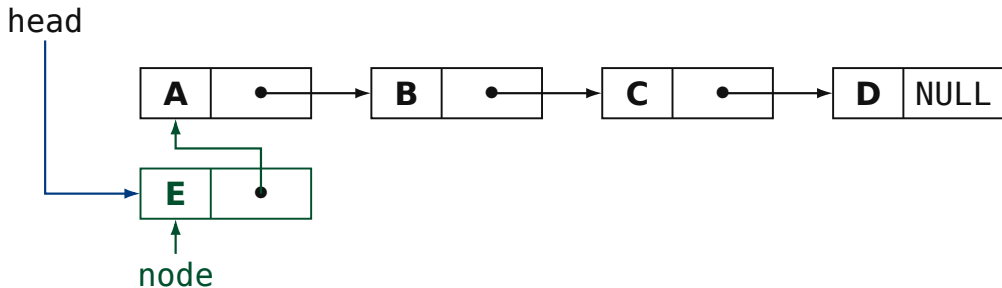


```
node->next = head;
```

- The old chain is now reachable through E.

Move head to E

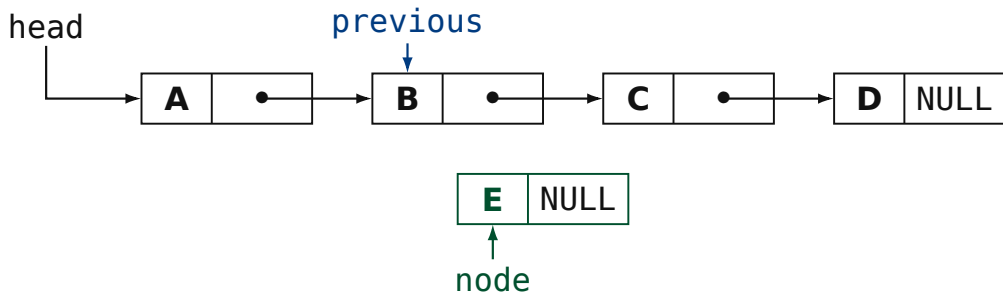
E already points to A. What must change now?



head = node;

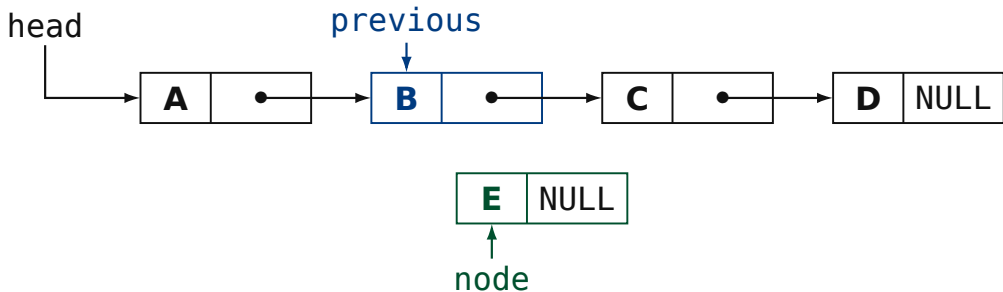
- **E → A → B → C → D.**

Insert in the middle · starting state



- Insert **E** between **B** and **C**. *previous* points to B.

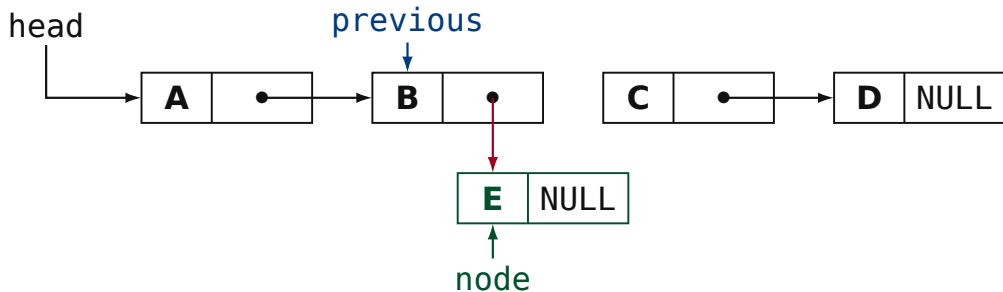
Insert in the middle · the link to preserve



- B's next link is currently our route to C and the rest of the chain.

If B points to E first

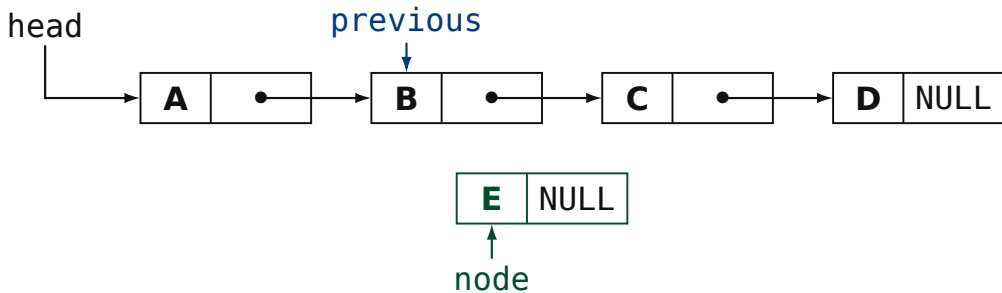
What remains reachable after changing B's link first?



```
previous->next = node;
```

- The list stops at E; **C and D leak**.

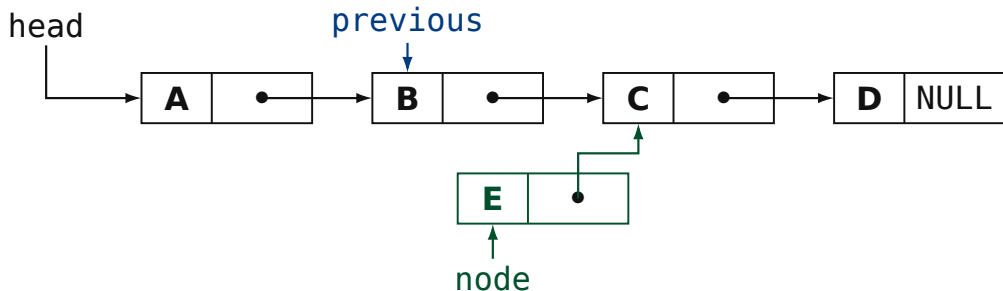
Back to the initial list



- Reset to $A \rightarrow B \rightarrow C \rightarrow D$ with E still separate.

Link E to C

Which address should E's *next* receive?

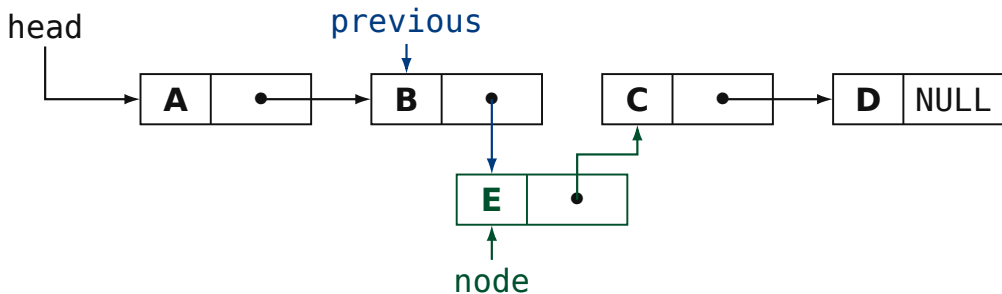


```
node->next = previous->next;
```

- E points to C; B still points to C.

Link B to E

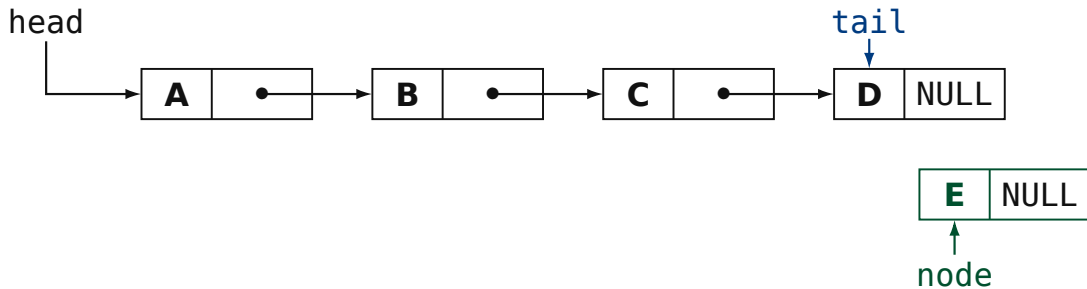
E already points to C. Which link finishes the insertion?



```
previous->next = node;
```

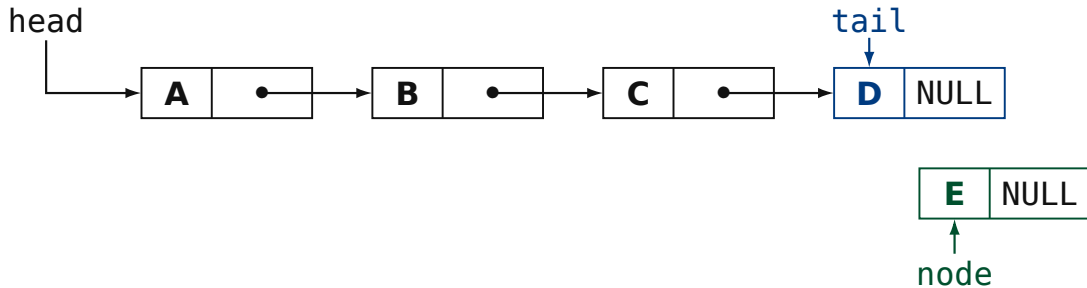
- The list is now **A → B → E → C → D**.

Insert at the tail · starting state



- Append **E** after **D**. *tail* points to D.

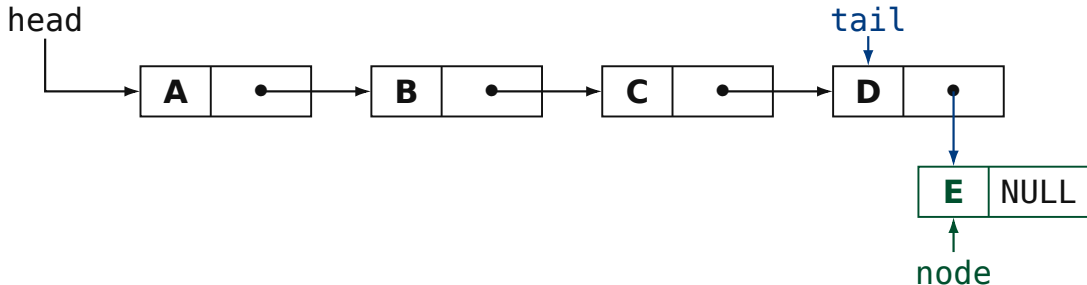
Insert at the tail · find the final link



- D's next link currently contains *NULL*.

Insert at the tail · link D to E

- Which link attaches E to the chain?

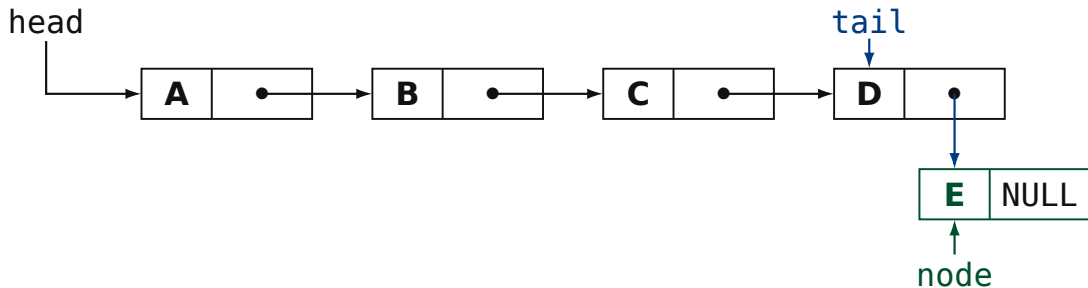


```
tail->next = node;
```

- Connect the old tail to the new node.

Insert at the tail · terminate the chain

What makes E the last node?



```
node->next = NULL;
```

- The list is now $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow \text{NULL}$.

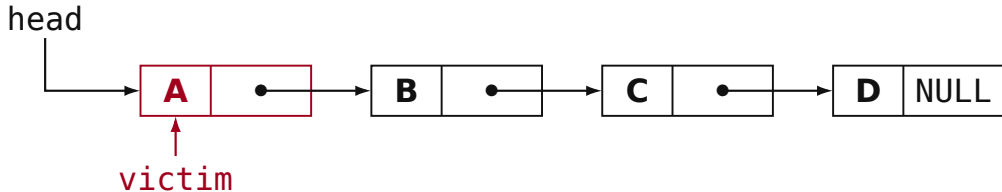
6. Unlink, then free

A song is removed from the playlist.

Reconnect the remaining songs and release its memory.

Delete the head · save the victim

- Which address must we keep before changing *head*?

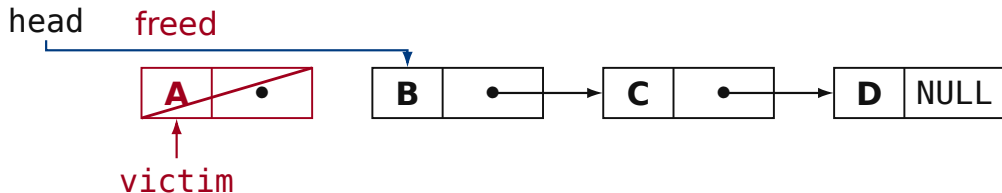


```
list_elem_t *victim = head;
```

- Keep the address of A so it can be freed after unlinking.

Delete the head · unlink, then release

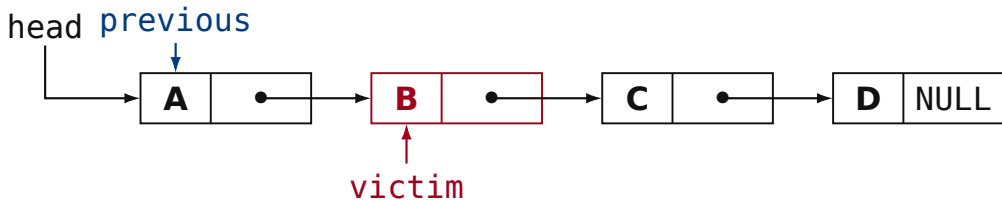
Where should *head* point before we release A?



```
head = victim->next;  
free(victim);
```

- B is now the first node. **Do not access A after free().**

Delete in the middle · save B

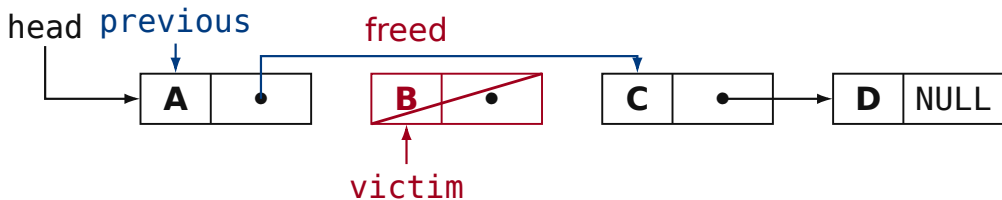


```
list_elem_t *victim = previous->next;
```

- *previous* points to A. Save the node to be removed.

Delete in the middle · reconnect A to C

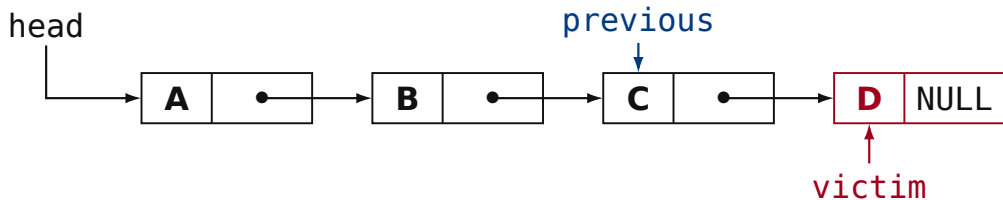
- How do we bypass B before releasing it?



```
previous->next = victim->next;  
free(victim);
```

- Preserve the successor** before freeing the removed node.

Delete the tail · find its predecessor

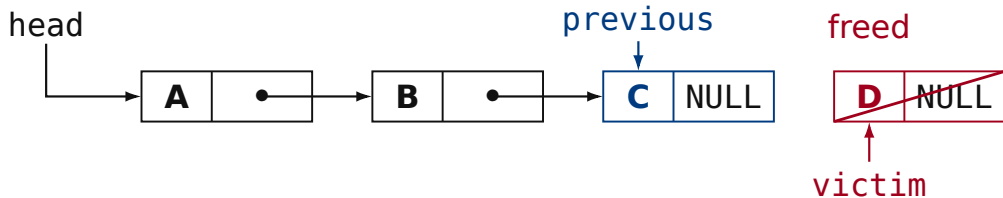


```
list_elem_t *victim = previous->next;
```

- *previous* points to C; save its successor D.

Delete the tail · make C the last node

What must C's next link become?



```
previous->next = NULL;  
free(victim);
```

- Set C's next link to *NULL*, then free D.

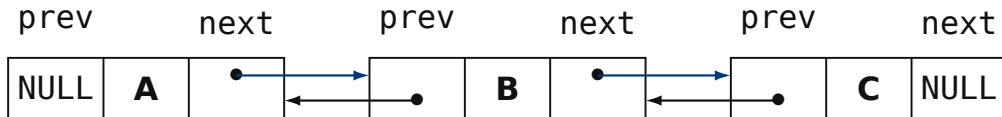
7. Doubly linked lists

Our player needs a **Previous** button.

How can we go back from the current song?

A link in each direction

What extra address should each node store?



- Add *prev* beside *next*: traverse in **both directions**.
- Unlink a known node without searching for its predecessor.
- **One extra pointer per node**; maintain **both links**.

Time and storage costs

n nodes; current node known, predecessor not saved.

Operation	Singly linked	Doubly linked
Find its predecessor	$O(n)$	$O(1)$
Unlink this node	$O(n)$	$O(1)$
Find node number k	$O(n)$ worst case	$O(n)$ worst case
Total node storage	$\Theta(n)$	$\Theta(n)$

- Singly linked: start at *head*, follow *next* to the predecessor.
- Doubly linked: read *prev* directly; **one extra pointer per node**.

8. Questions

Choose **A, B, C or D**, then explain why.

Linked lists · q1: changing the head

```
void prepend(list_elem_t *head,  
             list_elem_t *node) {  
    node->next = head;  
    head = node;  
}
```

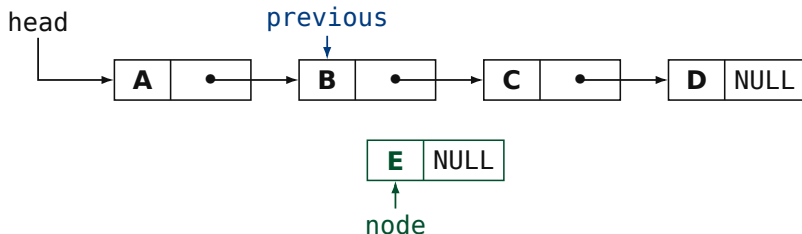
head points to A; *node* points to a separate node E.

After *prepend(head, node)*,
what happens to the caller's
head?

- A. It points to E.
 - B. **It still points to A.**
 - C. It becomes *NULL*.
 - D. It points to a copy of A.
-
- The function changes **its local pointer copy**.
 - E points to A, but the caller's *head* is unchanged.

Linked lists · q2: insertion order

Insert E between B and C. Which order works?



```
/* X */ previous->next = node;  
/* Y */ node->next = previous->next;
```

- A. X, then Y
- B. **Y, then X**
- C. Only X
- D. Only Y

- **E remembers C** before B points to E.

Linked lists · q3: releasing a node

victim points to the first node of a nonempty list.

```
free(victim);  
head = victim->next;
```

What is the problem?

- A. The whole list is freed.
 - B. *head* must become *NULL*.
 - C. **We read a freed node.**
 - D. There is no problem.
-
- Read *victim->next* **before free()**.
 - The invalid read has undefined behaviour; a crash is not guaranteed.

Linked lists · q4: time and storage

A doubly linked list has n nodes; the current node is known.

Reading *prev*: what time cost and **total** node storage?

A. $O(1)$ **time**; $\Theta(n)$ **storage**

B. $O(n)$ time; $\Theta(n)$ storage

C. $O(1)$ time; $\Theta(1)$ storage

D. $O(n)$ time; $\Theta(n^2)$ storage

- **Direct access** to the predecessor; no traversal.
- **One extra pointer per node**; total storage stays linear.